

Problem 7.3

The total amount of a loan to which interest has been added is \$20,000. The term of the loan was four and one-half years.

1. If money accumulated at simple interest at a rate of 6%, what was the amount of the loan?
2. If the annual rate of interest was 6% and interest was compounded annually, what was the amount of the loan?

Solution

Let $A = 20000$ be the accumulated amount, $r = 0.06$ the annual interest rate, and $t = 4.5$ years.

1. Simple interest

For simple interest,

$$A = P(1 + rt).$$

Solving for the principal P ,

$$P = \frac{A}{1 + rt} = \frac{20000}{1 + 0.06(4.5)} = \frac{20000}{1.27} \approx 15748.03.$$

Therefore, the amount of the loan was

$$\boxed{\$15,748.03}.$$

2. Compound interest, compounded annually

For compound interest,

$$A = P(1 + r)^t.$$

Solving for P ,

$$P = \frac{A}{(1 + r)^t} = \frac{20000}{(1.06)^{4.5}} \approx 15386.99.$$

Therefore, the amount of the loan was

$$\boxed{\$15,386.99}.$$